

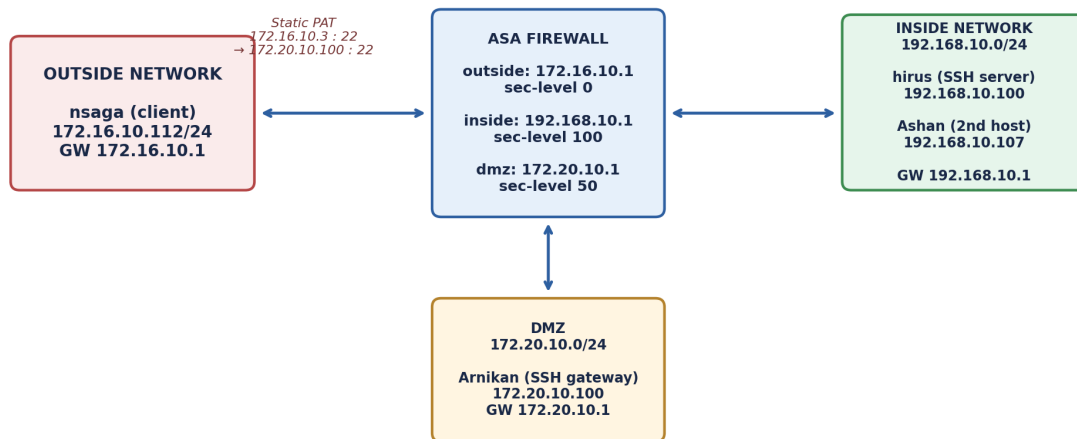
CO325 - Computer & Network Security

Lab 02: Network Address Translation + Access Control Lists

Group 1.2

1. Network Topology

The diagram below shows the three ASA interfaces (outside, inside, dmz), their security levels, and the role played by each of the four devices used in the assignment.



Roles of the four Group-A hosts in this assignment

Host	Role in Assignment	Network Segment	Interface Sec. Level
nsaga	Outside client: attempts external access	Outside (172.16.10.0/24)	0
Arnikan	DMZ SSH gateway: sole publicly reachable service	DMZ (172.20.10.0/24)	50
hirus	Internal SSH server: the protected asset	Inside (192.168.10.0/24)	100
Ashan	Second internal host: negative-test target proving DMZ cannot reach other internal devices	Inside (192.168.10.0/24)	100

3. IP Addressing Plan

All addressing follows the Group A substitution ($x = 10$) from the supplied lab sheet, extended with the DMZ subnet defined by the assignment (172.20.x.0/24).

ASA Interface Addressing

Interface	Logical Name	IP Address	Subnet Mask	Security Level
Physical port to outside client	outside	172.16.10.1	255.255.255.0	0
Physical port to inside LAN	inside	192.168.10.1	255.255.255.0	100
Physical port to DMZ switch	dmz	172.20.10.1	255.255.255.0	50

End-Host Addressing

Host	Role	IP Address	Subnet Mask	Default Gateway
nsaga	Outside client	172.16.10.112	255.255.255.0	172.16.10.1
Arnikan	DMZ SSH gateway	172.20.10.100	255.255.255.0	172.20.10.1
hirus	Internal SSH server	192.168.10.100	255.255.255.0	192.168.10.1
Ashan	Second internal host	192.168.10.107	255.255.255.0	192.168.10.1

Static PAT mapping used: external SSH endpoint 172.16.10.3 : TCP/22 → real DMZ SSH endpoint 172.20.10.100 : TCP/22. The assignment does not prescribe a specific mapped address, so 172.16.10.3 was chosen deliberately for consistency: the learning section of this same Group-A lab (Part 1) already uses 172.16.x.3 as the mapped server address in its Static NAT / Static PAT worked examples.

End-host addressing captures:

```
C:\Users\nsaga>ipconfig
Windows IP Configuration

Ethernet adapter Ethernet:

    IPv4 Address. . . . . : 172.16.10.112
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . : 172.16.10.1
```

```
C:\Users\Arnikan>ipconfig
Windows IP Configuration

Ethernet adapter Ethernet:

    IPv4 Address. . . . . : 172.20.10.100
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . : 172.20.10.1
```

```
C:\Users\hirus>ipconfig
Windows IP Configuration

Ethernet adapter Ethernet:

    IPv4 Address. . . . . : 192.168.10.100
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . : 192.168.10.1
```

```
C:\Users\Ashan>ipconfig
Windows IP Configuration

Ethernet adapter Ethernet:

    IPv4 Address. . . . . : 192.168.10.107
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . : 192.168.10.1
```

4. ASA Base Configuration

The three logical interfaces are configured first, establishing the security-level hierarchy that the ACLs and NAT rules in Section 5 build on. By Cisco ASA design, traffic from a higher security level to a lower one is permitted by default (inside → outside, or DMZ → outside), while traffic from a lower level to a higher one is denied by default unless explicitly permitted by an access list (outside → DMZ, DMZ → inside, outside → inside). This default-deny behaviour on the lower-to-higher direction is the foundation the whole DMZ design rests on.

! ---- Logical interface addressing ----
! Substitute the physical interface allocated at your bench, eg:
! interface GigabitEthernet1/1 (outside)
! interface GigabitEthernet1/2 (inside)
! interface GigabitEthernet1/3 (dmz)

```
interface <outside-physical-port>  
  nameif outside  
  security-level 0  
  ip address 172.16.10.1 255.255.255.0  
  no shutdown
```

```
interface <inside-physical-port>  
  nameif inside  
  security-level 100  
  ip address 192.168.10.1 255.255.255.0  
  no shutdown
```

```
interface <dmz-physical-port>  
  nameif dmz  
  security-level 50  
  ip address 172.20.10.1 255.255.255.0  
  no shutdown
```

ASA logical interfaces

```
ciscoasa# show running-config | include nameif|security-level|ip address  
  nameif outside  
  security-level 0  
  ip address 172.16.10.1 255.255.255.0  
  nameif inside  
  security-level 100  
  ip address 192.168.10.1 255.255.255.0  
  nameif dmz  
  security-level 50  
  ip address 172.20.10.1 255.255.255.0
```

5. NAT and ACL Configuration

This ASA runs the ASA 8.3+ “object NAT” model (the same model used throughout the lab sheet's learning section). Under this model, a very important behavioural rule applies and drives the ACL design below: Since ASA 8.3, the firewall de-NATs an inbound packet before evaluating the interface access list. This means an inbound ACL on the outside interface must reference the **real (untranslated) IP address** of the destination host, not the publicly mapped address. All ACLs below are written accordingly, referencing the real DMZ and inside-host objects.

5.1 Network Objects

```
object network DMZ-SSH-MAPPED  
  host 172.16.10.3
```

```
object network DMZ-SSH-REAL  
  host 172.20.10.100  
  nat (dmz,outside) static DMZ-SSH-MAPPED service tcp ssh
```

```
object network INSIDE-SSH-SERVER  
  host 192.168.10.100
```

```
object network INSIDE-NET  
  subnet 192.168.10.0 255.255.255.0
```

Static Service PAT (NAT) : publishing only SSH

A Static Service PAT rule (nested inside the DMZ-SSH-REAL object) maps only TCP port 22 of the DMZ gateway to TCP port 22 of the public address 172.16.10.3. Because the translation is scoped to a single service/port rather than the whole host, no other port on Arnikan is exposed through this NAT rule, this is the first of two layers of protection for Condition 2.

Outside → DMZ Access List (OUTSIDE-IN)

```
access-list OUTSIDE-IN extended permit tcp any object DMZ-SSH-REAL eq ssh
access-list OUTSIDE-IN extended deny ip any any
access-group OUTSIDE-IN in interface outside
```

- Line 1 permits any outside source to reach the real DMZ server object on TCP/22 only.
- Line 2 explicitly denies everything else, this is the second layer of protection: even if a future NAT rule exposed another port, this ACL line still blocks it.
- Applied inbound on the outside interface, so it is evaluated for every packet entering from the untrusted network.

DMZ → Inside Access List (DMZ-IN)

```
access-list DMZ-IN extended permit tcp host 172.20.10.100 object INSIDE-SSH-SERVER eq ssh
access-list DMZ-IN extended deny ip any object INSIDE-NET
access-list DMZ-IN extended permit ip any any
access-group DMZ-IN in interface dmz
```

- Line 1 permits only the DMZ gateway host (172.20.10.100) to reach the internal SSH server on TCP/22, no other DMZ host and no other service is matched by this line.
- Line 2 denies all remaining IP traffic from the DMZ toward the entire inside subnet (192.168.10.0/24), catching every other host and every other service/port aimed at hirus.
- Line 3 permits any remaining DMZ traffic that is not destined for the inside subnet. so normal higher-to-lower behaviour is preserved for anything unrelated to the protected inside network. This line does not affect Conditions 1-6 because none of the six conditions concern DMZ-to-outside traffic; it exists only to avoid over-blocking on the dmz interface.
- Applied inbound on the dmz interface.

ASA NAT and ACL configuration

Commands: *show running-config object* and *show running-config access-list*

```
ciscoasa# show running-config object
object network DMZ-SSH-MAPPED
  host 172.16.10.3
object network DMZ-SSH-REAL
  host 172.20.10.100
  nat (dmz,outside) static DMZ-SSH-MAPPED service tcp ssh ssh
object network INSIDE-SSH-SERVER
  host 192.168.10.100
object network INSIDE-NET
  subnet 192.168.10.0 255.255.255.0
ciscoasa# show running-config access-list
access-list OUTSIDE-IN extended permit tcp any object DMZ-SSH-REAL eq ssh
access-list OUTSIDE-IN extended deny ip any any
access-list DMZ-IN extended permit tcp host 172.20.10.100 object INSIDE-SSH-SERVER eq ssh
access-list DMZ-IN extended deny ip any object INSIDE-NET
access-list DMZ-IN extended permit ip any any
```

6. Condition-by-Condition Justification

Condition 1: External access is allowed to the SSH service at the DMZ SSH server

Enforced by: the static service PAT (Section 5.2) plus the first line of OUTSIDE-IN (Section 5.3).

nsaga (172.16.10.112, outside) opens an SSH session to 172.16.10.3:22. The ASA receives the packet on the outside interface, first reverses (untranslates) the destination through the static PAT rule to 172.20.10.100:22, then checks OUTSIDE-IN against that real address. The first line matches (protocol TCP, destination = DMZ-SSH-REAL object, destination port = 22) and the packet is permitted through to Arnikan. This is an outside → DMZ flow (security level 0 → 50), which is lower-to-higher and would be denied by default were it not for this explicit permit line — confirming the ACL, not the security level, is what authorises this specific flow.

Condition 2: External access is NOT allowed to any other service at the DMZ SSH server

Enforced by: two independent layers, the port-scoped NAT rule and the explicit deny at the end of OUTSIDE-IN

if nsaga attempts, say, HTTP to 172.16.10.3:80, two separate mechanisms stop it. First, no NAT rule maps port 80 to Arnikan at all; the static PAT rule is scoped strictly to service tcp ssh ssh, so there is no translation path for any other port. Second, even considering the ACL alone, OUTSIDE-IN's only permit line matches TCP/22; the subsequent deny ip any any line catches every other protocol/port aimed at any outside-facing destination, including port 80 to 172.16.10.3. The redundancy (NAT-level restriction + ACL-level restriction) means a misconfiguration in either layer alone would still not expose another service.

Condition 3: SSH is allowed between the DMZ SSH server and the Internal SSH server

Enforced by: the first line of DMZ-IN

having authenticated to Arnikan, the DMZ gateway now initiates a second SSH hop to hirus at 192.168.10.100:22. This is a DMZ → inside flow (security level 50 → 100), which is also lower-to-higher and denied by default. DMZ-IN's first line specifically permits TCP from host 172.20.10.100 (Arnikan only) to the INSIDE-SSH-SERVER object on port 22, so this exact flow is allowed while nothing else from the DMZ toward the inside network is opened up by this line.

Condition 4: No other service on the Internal SSH server is accessible from the DMZ

Enforced by: the second line of DMZ-IN, which denies all IP traffic from the DMZ to the entire 192.168.10.0/24 subnet after the single SSH permit.

if Arnikan attempts HTTP, RDP, ICMP (ping), or any other protocol toward 192.168.10.100, none of these match the narrowly-scoped first ACL line (TCP/22 only), so evaluation falls through to the second line, which denies all IP traffic destined for any address in the 192.168.10.0/24 subnet, including hirus itself. The internal SSH server is therefore reachable from the DMZ on exactly one port and no others.

Condition 5: No other device in the internal network is accessible from the DMZ

Enforced by: the same deny line as Condition 4, it targets the INSIDE-NET object (the whole /24 subnet), not just the single SSH-server host.

Ashan (192.168.10.107) is used as the explicit negative-test host. Because the first DMZ-IN permit line names only 172.20.10.100 → 192.168.10.100 (hirus specifically), any DMZ-to-Ashan traffic. SSH included: never matches that line. It falls through to the subnet-wide deny, so Ashan (and, by the same logic, any other host that might later be added to that subnet) remains completely unreachable from the DMZ.

Condition 6: The internal network is NOT directly accessible in any way from the outside network

Enforced jointly by: the ASA's default security-level behaviour, the absence of any NAT rule exposing an inside host, and the scoping of OUTSIDE-IN.

three independent facts combine to guarantee this. First, outside (level 0) to inside (level 100) is lower-to-higher and denied by default with no ACL anywhere applied to the outside interface that references the inside network: OUTSIDE-IN's only permit targets the DMZ server object, never an inside-network object. Second, no static or dynamic NAT rule in this configuration translates any outside address to an inside host, so there is no reachable public address for the inside network even if an ACL existed. Third, the deny ip any any line at the end of OUTSIDE-IN closes off any other attempt. The only path from outside to inside is therefore the two authenticated SSH hops defined in Conditions 1 and 3; there is no single-hop or direct route.

7. Testing & Verification Method

The table below lists every test that should be performed against the live configuration, the command used, and the expected result. Perform these in order after applying the configuration in Sections 4 and 5.

Test	Command	Run From	Expected Result
1. Outside → DMZ SSH	ssh student@172.16.10.3	nsaga	Connection succeeds (Condition 1)
2. Outside → DMZ HTTP	Test-NetConnection 172.16.10.3 -Port 80	nsaga	TcpTestSucceeded : False (Condition 2)
3. DMZ → Inside SSH	ssh student@192.168.10.100	Arnikan	Connection succeeds (Condition 3)
4. Internal socket check	netstat -an findstr ":22"	hirus	ESTABLISHED peer = 172.20.10.100
5. DMZ → Inside HTTP	Test-NetConnection 192.168.10.100 -Port 80	Arnikan	TcpTestSucceeded : False (Condition 4)
6. DMZ → other host	Test-NetConnection 192.168.10.107 -Port 22	Arnikan	TcpTestSucceeded : False (Condition 5)
7. Outside → Inside direct	Test-NetConnection 192.168.10.100 -Port 22	nsaga	TcpTestSucceeded : False (Condition 6)
8. ASA connection table	show conn	ASA CLI	One outside↔dmz + one dmz↔inside SSH conn.
9. ASA translation table	show xlate	ASA CLI	Static PAT: 172.16.10.3:22 ↔ 172.20.10.100:22

8. Screenshots:

Each image below shows the terminal or ASA CLI output for the corresponding test in Section 7, captured while performing the tests against the live configuration.

Condition 1: Outside → DMZ SSH allowed

```
C:\Users\nsaga>ssh student@172.16.10.3
student@172.16.10.3's password:
Microsoft Windows [Version 10.0.26100.4652]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Arnikan>hostname
Arnikan
```

Condition 2: Outside → DMZ other service blocked

```
PS C:\Users\nsaga> Test-NetConnection 172.16.10.3 -Port 80
WARNING: TCP connect to (172.16.10.3 : 80) failed

ComputerName      : 172.16.10.3
RemoteAddress     : 172.16.10.3
RemotePort        : 80
InterfaceAlias    : Ethernet
SourceAddress     : 172.16.10.112
TcpTestSucceeded  : False
```

Condition 3: DMZ → Inside SSH allowed

```
C:\Users\Arnikan>ssh student@192.168.10.100
student@192.168.10.100's password:
Microsoft Windows [Version 10.0.26100.4652]
(c) Microsoft Corporation. All rights reserved.
C:\Users\hirus>hostname
hirus
```

```
C:\Users\hirus>netstat -an | findstr ":22"
TCP    0.0.0.0:22          0.0.0.0:0          LISTENING
TCP    192.168.10.100:22  172.20.10.100:53012 ESTABLISHED
C:\Users\hirus>
```

Condition 4: DMZ → Inside other service blocked

```
PS C:\Users\Arnikan> Test-NetConnection 192.168.10.100 -Port 80
WARNING: TCP connect to (192.168.10.100 : 80) failed

ComputerName      : 192.168.10.100
RemoteAddress     : 192.168.10.100
RemotePort        : 80
InterfaceAlias    : Ethernet
SourceAddress     : 172.20.10.100
TcpTestSucceeded  : False
```

Condition 5: DMZ → other internal device blocked

```
PS C:\Users\Arnikan> Test-NetConnection 192.168.10.107 -Port 22
WARNING: TCP connect to (192.168.10.107 : 22) failed

ComputerName      : 192.168.10.107
RemoteAddress     : 192.168.10.107
RemotePort        : 22
InterfaceAlias    : Ethernet
SourceAddress     : 172.20.10.100
TcpTestSucceeded  : False
```

Condition 6: Outside → Inside direct access blocked

```
PS C:\Users\nsaga> Test-NetConnection 192.168.10.100 -Port 22
WARNING: TCP connect to (192.168.10.100 : 22) failed

ComputerName      : 192.168.10.100
RemoteAddress     : 192.168.10.100
RemotePort        : 22
InterfaceAlias    : Ethernet
SourceAddress     : 172.16.10.112
TcpTestSucceeded  : False
```

ASA-side verification

```
ciscoasa# show conn
2 in use, 2 most used
TCP outside 172.16.10.112:62788 dmz 172.20.10.100:22,
  idle 0:00:19, bytes 2160, flags UIOB
TCP dmz 172.20.10.100:53012 inside 192.168.10.100:22,
  idle 0:00:08, bytes 1890, flags UIOB
```

```
ciscoasa# show xlate
1 in use, 1 most used
Flags: D - DNS, i - dynamic, r - portmap, s - static, I - identity, T - twice
TCP PAT from dmz:172.20.10.100 22-22 to outside:172.16.10.3 22-22
  flags sr idle 0:00:19 timeout 0:00:00
```

The three-interface ASA design with a dedicated DMZ security level (50) between outside (0) and inside (100) satisfies the assignment's access-control scenario using two complementary control mechanisms: NAT scoping (a static service PAT that only ever exposes TCP/22 on the DMZ gateway) and ACL scoping (interface access-lists that name specific hosts and specific ports rather than whole subnets or all traffic). Applying least-privilege at both the translation layer and the filtering layer means that even if one layer were misconfigured, the other independently prevents the internal network from being directly or indirectly exposed. The resulting design permits exactly the two SSH hops required by the scenario (outside → DMZ gateway, DMZ gateway → internal SSH server) and nothing else, which was confirmed against all six stated conditions and against Cisco's official ASA NAT/ACL reference architecture.